1 An object of mass 0.050 kg is projected from the top of a 50 m tall building at an angle of 60° to the horizontal with an initial speed of 20 m s^{-1} as shown in Fig. 1.1.

A is a point in the trajectory that is at the same height as the object initially and *d* is the horizontal distance the object lands from the building.

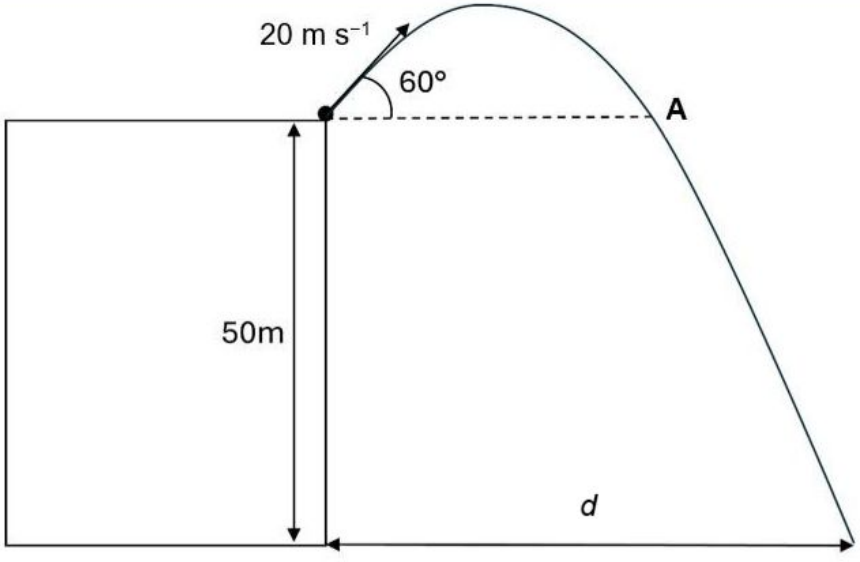


Fig. 1.1

(a) Assuming there is no resistance,

(i) Calculate the distance *d*.

taking the upward direction as positive,

vertical component of initial velocity, $u_y = 20\sin 60°$ (M1)

let t to be the time taken to reach the ground,

$s_y = u_y t + \frac{1}{2} a_y t^2$

$-50 = 20\sin 60° t - 4.905t^2$ (M1)

$t = 5.414$ s

$d = u_x t = (20\cos 60°)(5.414) = 54.1$ m (A1)

d = ... m [3]

(ii) Without calculation, state the speed of the object at **A** and explain how you arrived at your answer.

$20\ \text{m s}^{-1}$ [B1]

As the projectile rises to its maximum height, some of its kinetic energy is converted into gravitational potential energy. As it falls from the maximum height to point A (which is at the same height as the point of projection), the same amount of gravitational potential energy is converted back into kinetic energy.

Since mechanical energy is conserved (neglecting air resistance), the kinetic energy at point A is equal to the initial kinetic energy. Hence, the speed at A is the same as the initial speed. [B1]

OR

The projectile is at the same height initially and at point A, so its gravitational potential energy is the same at both positions.

As mechanical energy is conserved (neglecting air resistance), its kinetic energy must also be the same at these two positions. Therefore, its speed at A is equal to its initial speed. [B1]

..... [2]

(b) Experimentally, it is found that the same object now reaches a maximum height of 60.5 m above the ground in its trajectory.

(i) Show quantitatively that the air resistance is not negligible in this case.

neglecting the air resistance,
the maximum height s_y above the point of projection is given by:

$v_y^2 = u_y^2 + 2a_y s_y$

$0 = (20\sin 60°)^2 + 2(-9.81)s_y$

$s_y = 15.2905\ \text{m} = 15.3\ \text{m}$ (to 3 s.f.)

$\Rightarrow$ the maximum height above the building top $= 15.3$ m (M1)

factoring in air resistance,
the maximum height above the building top $= 10.5$ m

since $15.3\ \text{m} > 10.5\ \text{m}$, (M1)
the air resistance is not negligible in the stated case

[2]

(ii) The speed of the object at its maximum height is 8.0 m s^{-1}.
Calculate the energy lost due to air resistance at maximum height.

initial kinetic energy, $KE_i = \frac{1}{2}mu^2 = 10.0$ J

kinetic energy at its maximum height, $KE_f = \frac{1}{2}mv^2 = 1.6$ J

gain in gravitational potential energy,
$\Delta GPE = mgs_y = (0.050)(9.81)(10.5) = 5.15$ J (M1)

By conservation of energy,
$KE_i = KE_f + \Delta GPE +$ energy lost due to air resistance
energy lost $= (KE_i - KE_f) - \Delta GPE = (10.0 - 1.6) - 5.15 = 3.25$ J (A1)

energy lost = ... J [2]

(c) Sketch the path of the object in Fig. 1.1 when air resistance is not negligible. [1]

[Total: 10]